

Introduction to Bayesian Statistics

Part 3 Prior & Posterior Distributions

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iDiv 2026



This lecture

Short summary of last lecture

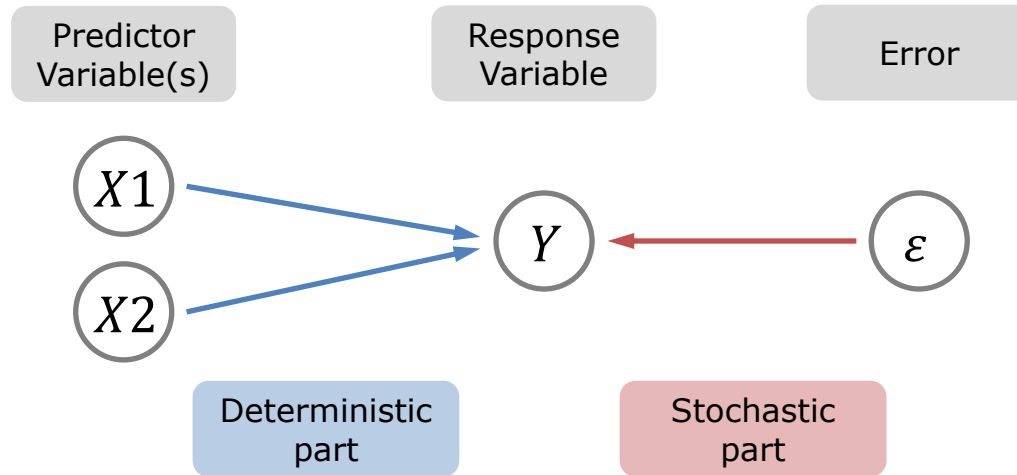
Some useful distributions

The prior distribution

The posterior distribution

Posterior predictions and model evaluation

From last lecture



From last lecture

Bayes' rule

The diagram shows the Bayes' rule equation $p(\theta|y) = \frac{p(y|\theta) \cdot p(\theta)}{p(y)}$ inside a light beige rounded rectangle. Four arrows point to the components of the equation: 'Likelihood function $L(\theta)$ ' points to $p(y|\theta)$, 'Prior distribution' points to $p(\theta)$, 'Posterior distribution' points to $p(\theta|y)$, and 'Normalization constant Independent of θ ' points to $p(y)$.

$$p(\theta|y) = \frac{p(y|\theta) \cdot p(\theta)}{p(y)}$$

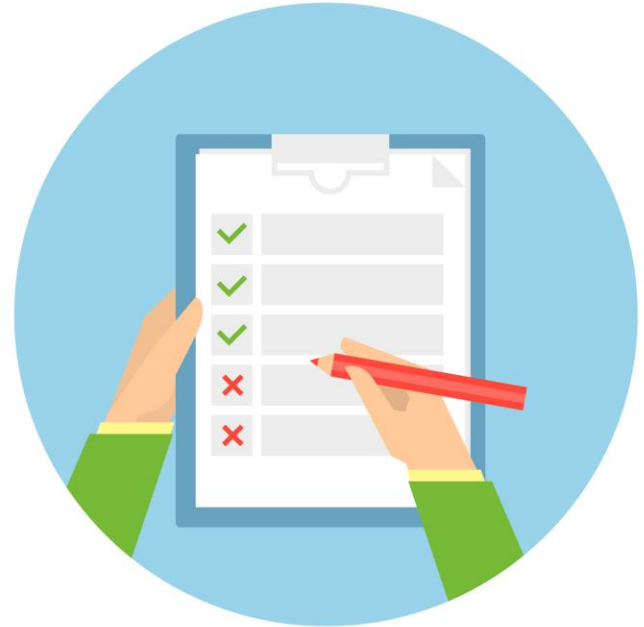
Update beliefs (prior)
by gaining new information (data & likelihood)

Posterior distribution used for
quantitative & direct statements on research questions

Don't have access to posterior distribution
Approximate by MCMC sampling

From last lecture

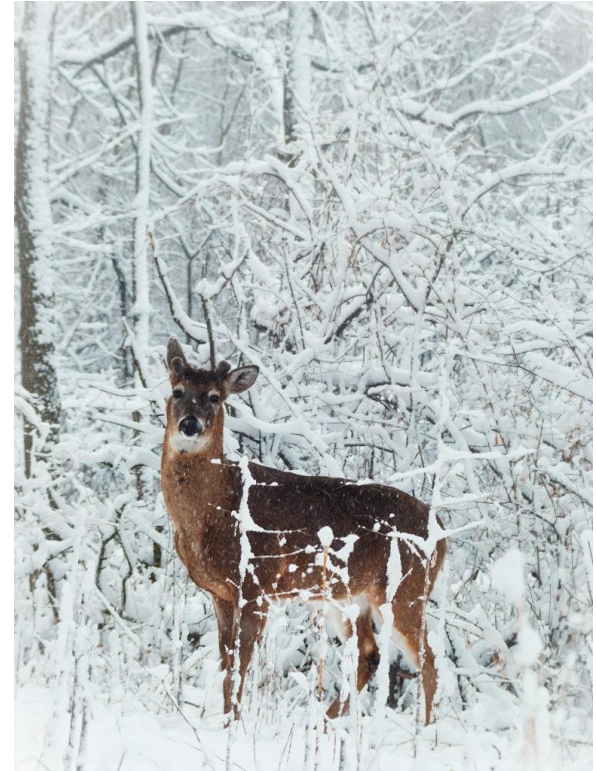
- 1) Research question (hypotheses)
 - 2) Data collection
 - 3) Statistical model
 - 4) Prior distribution choice
 - 5) Model fitting (MCMC)
 - 6) Evaluate model output
 - 7) Quantitative statements on hypotheses
- Revise model



From last lecture

- **Example:** number of individuals from a population of $N = 10$ that survive the winter
- y **discrete** and **bounded** variable with outcomes $0, 1, 2, \dots, 10$
- Average survival probability $\theta = 0.6$ (60%)
- Binomial distribution: $y \sim \text{Binomial}(N, \theta)$

random variable „distributed as” parameters:
size N
probability θ



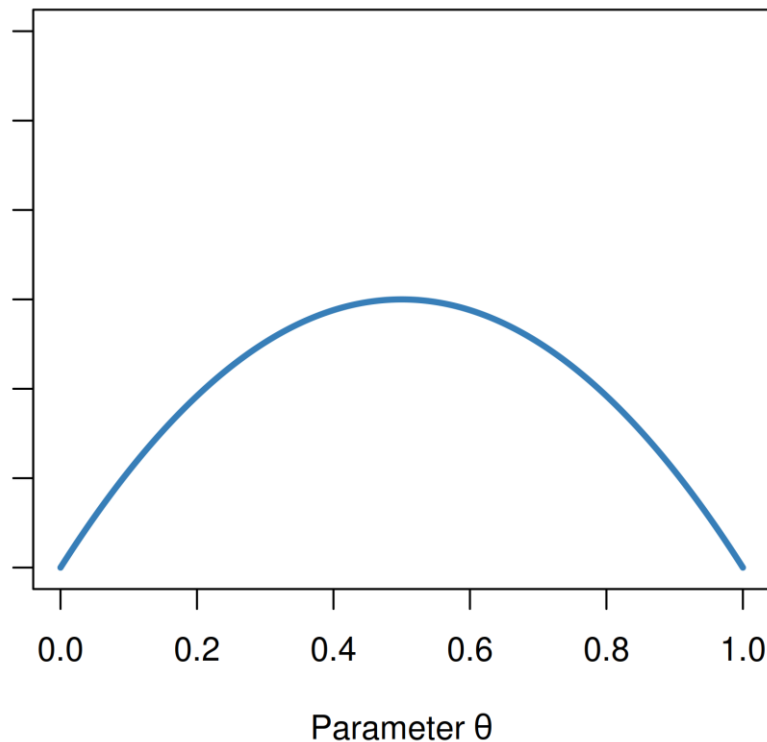
From last lecture

Prior distribution

Chosen by **you**

Density known over full parameter range

$$p(\theta) = \text{dbeta}(\theta \mid 2, 2)$$



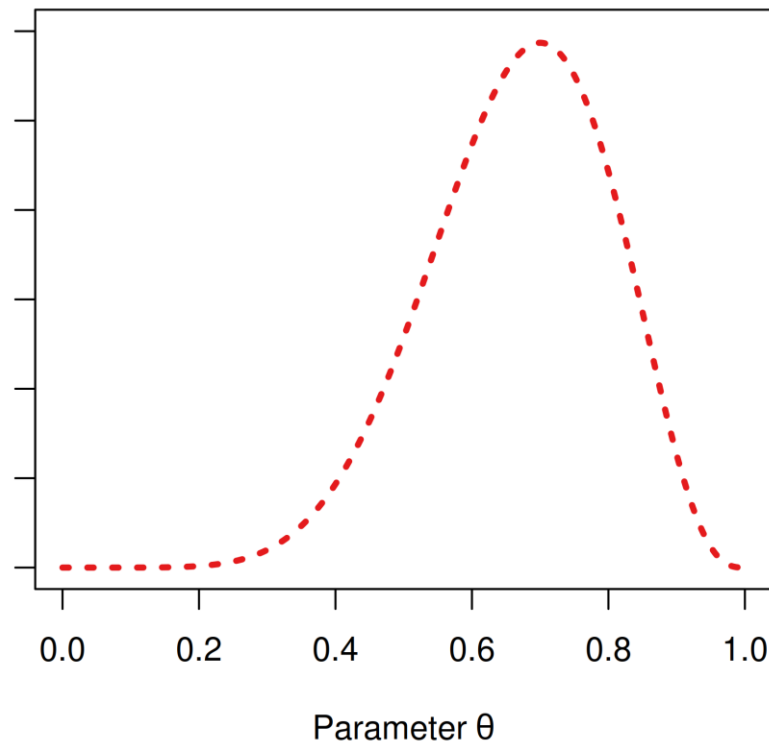
From last lecture

Likelihood function

Defined by **your** data and **your** statistical model
(deterministic & stochastic part)

Can be computed for every single parameter value
But values not known over full parameter range

$$\begin{aligned} L(\theta) &= \prod_{i=1}^n p(y_i | \theta) \\ &= \prod_{i=1}^n \text{dBinom}(\text{survived}_i, \text{total}_i | \theta) \end{aligned}$$



From last lecture

Posterior distribution

Software **output**:
approximation by MCMC sampling

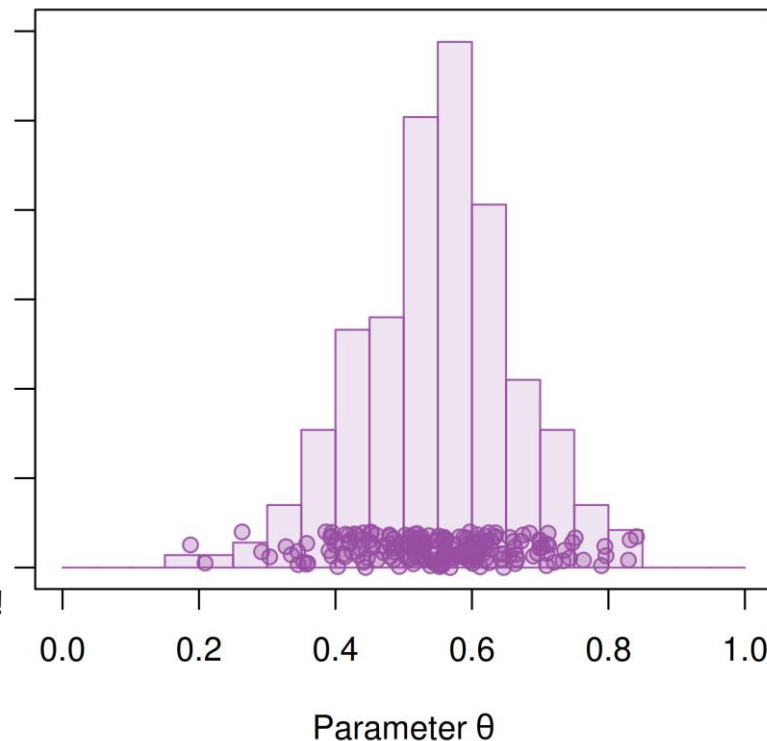
$$p(\theta|y) \sim p(\theta) \cdot L(\theta)$$

Represented by samples $\theta_1, \theta_2, \theta_3, \dots, \theta_{1000}$ only!

Posterior density values $p(\theta_1|y), p(\theta_2|y), \dots, p(\theta_{1000}|y)$
not required, don't need to be saved

Counting instead of integrating for hypotheses testing!

Super easy once
you have a
posterior sample



Distribution zoo app

<https://ben18785.shinyapps.io/distribution-zoo/>

Ben Lambert and Fergus Cooper

Last month: used by 203 people over 408 sessions in 33 countries

Since created: used by 19498 people over 36776 sessions in 144 countries

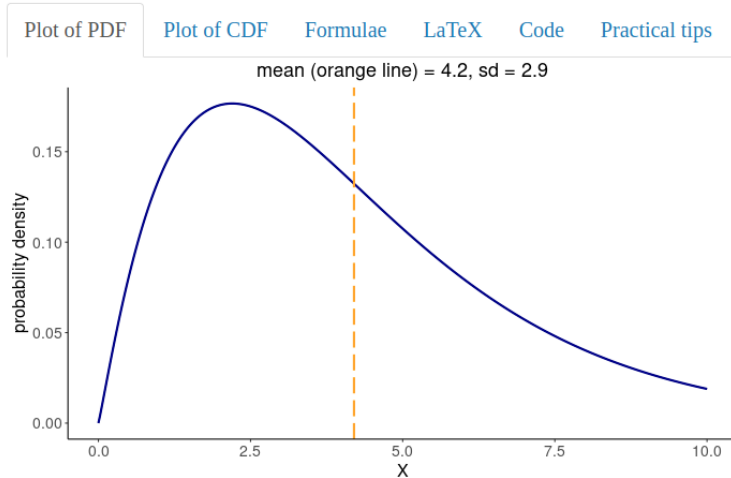
Category of distribution
Continuous Univariate

Distribution type:
Gamma

Range
10 1,000

Shape parameter
0.1 2.1 10

Rate parameter
0.1 0.5 2



Probability playground app

<https://www.acsu.buffalo.edu/~adamcunn/probability/probability.html>

The gamma distribution is a "waiting time" distribution. Suppose events occur independently and randomly with an average time between events of β . The waiting time until α events have occurred is a $\text{gamma}(\alpha, \beta)$ random variable.

The parameter α is known as the shape parameter, and the parameter β is called the scale parameter. Increasing α leads to a more "peaked" distribution, while increasing β increases the "spread" of the distribution.

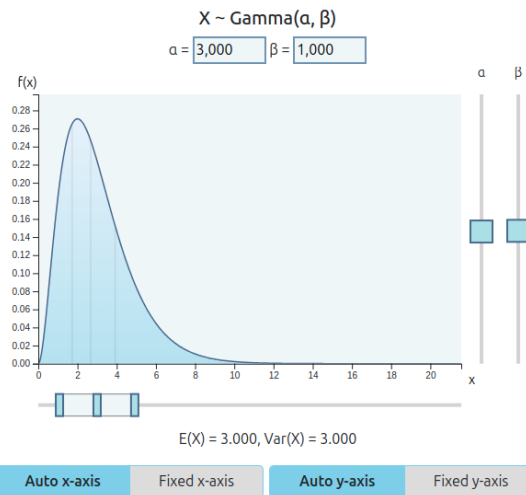
The function $\Gamma(s)$ in the denominator of the pdf and cdf denotes the gamma function, while the function $\gamma(s, x)$ in the cdf denotes the lower incomplete gamma function.

Parameter	Range	Description
α	$\alpha > 0$	Shape parameter
β	$\beta > 0$	Scale parameter

Probability Density Function	Support
$f(x; \alpha, \beta) = \frac{1}{\Gamma(\alpha)\beta^\alpha} x^{\alpha-1} e^{-x/\beta}$	$0 \leq x < \infty$

Mean	Variance
$\alpha\beta$	$\alpha\beta^2$

Example	α	β
A radioactive substance emits two alpha particles every second on average. Let X be the waiting time for three particles to be emitted.	3.000	0.5000
Cars arrive at an intersection at an average rate of one every two minutes. Let X be the waiting time until five cars have arrived.	5.000	2.000
Garage door lightbulbs last five years on average and are replaced when they fail. Let X be the time that a box of six bulbs lasts.	6.000	5.000



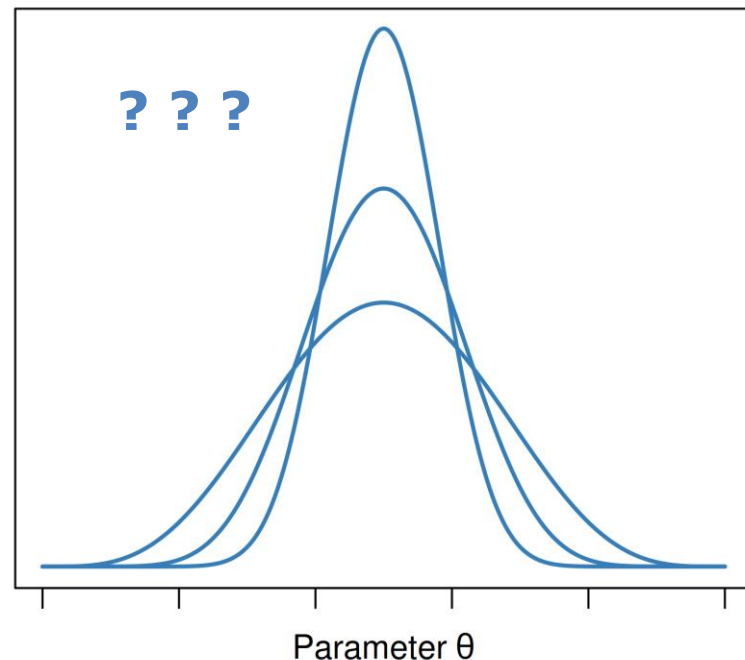
Note that the mean $\alpha\beta$ is directly proportional to both α and β . This is what we would intuitively expect - the mean time spent waiting for α events to occur increases in proportion to both the number of events α and the average time β between events.

The shape of the pdf depends on the parameter α . For values of $\alpha \leq 1$, the pdf is strictly decreasing. For values of $\alpha > 1$, the pdf is unimodal.

Prior distributions

Prior information

- Priors represent belief about model parameters (for example effect size of an x-y association)
- *Traditional* viewpoint: before we see the data y
Data information is already contained in the likelihood!
- Use information from
 - General expectation / reasonable range
 - Previous experiments
 - Related studies in the literature
- *Modern* viewpoint:
 - Priors used for regularization
 - Prior predictive checks:
Are predictions from the prior in the same range / magnitude as observed data?
- Priors are problem-specific



Types of priors

Flat / uninformative prior

- You know absolutely nothing about the parameter
- This is rarely the case

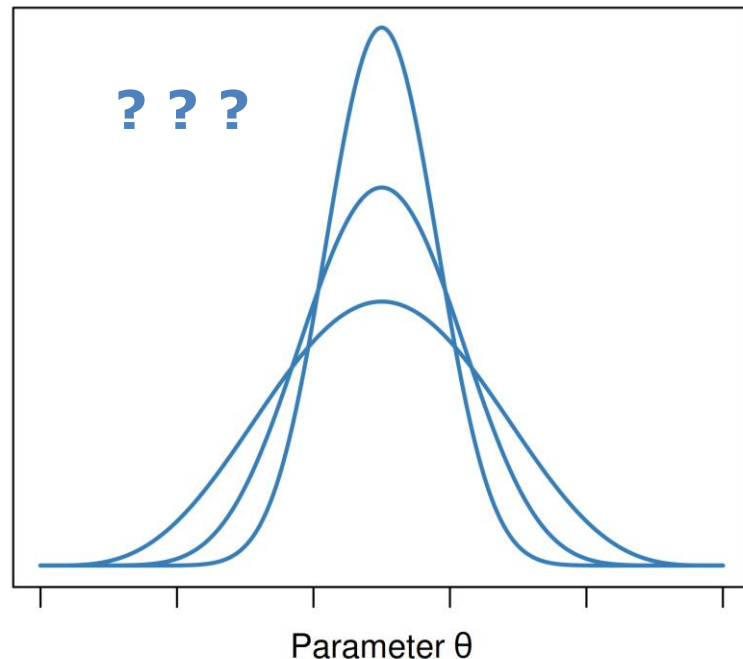
Vague / weakly informative prior

- You have a vague idea
- For example about the order of magnitude, or sign

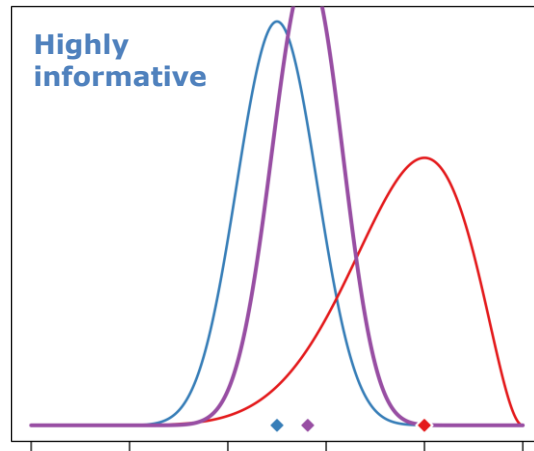
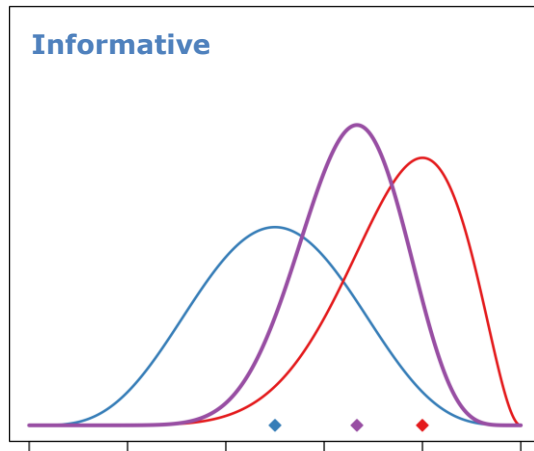
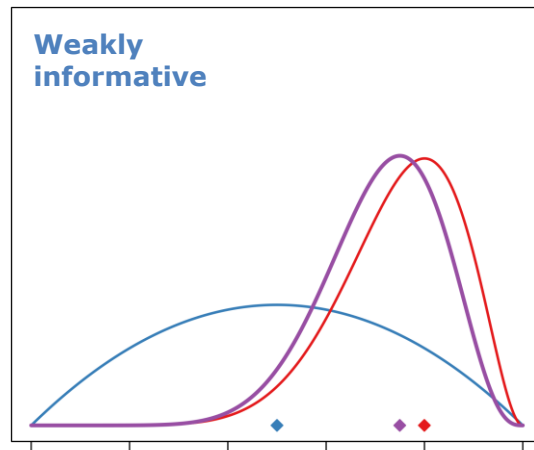
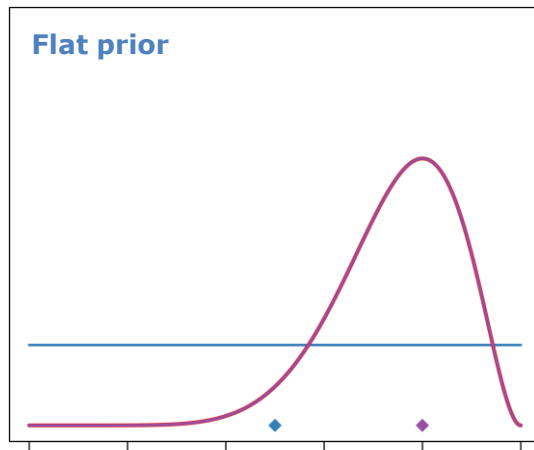
Informative prior

- You have some idea about the parameter

→ There is no formal definition of these terms!



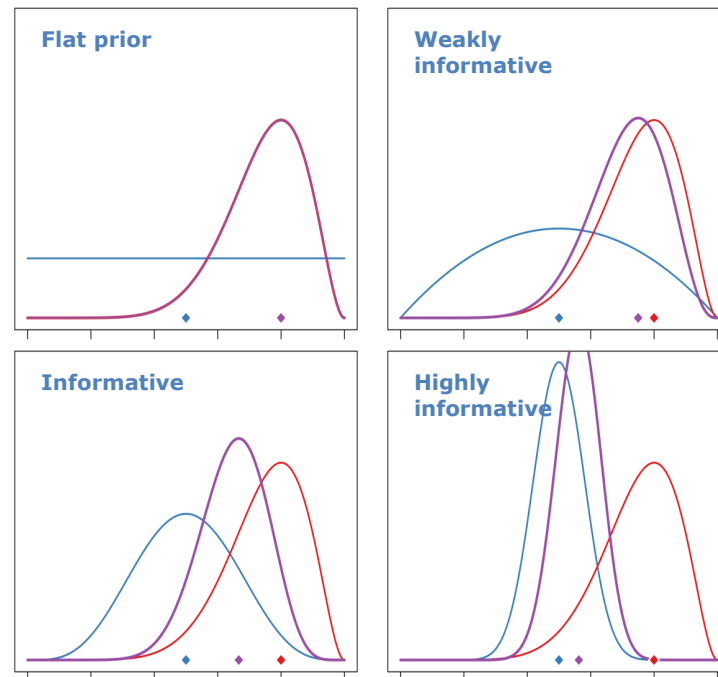
Prior affects the posterior



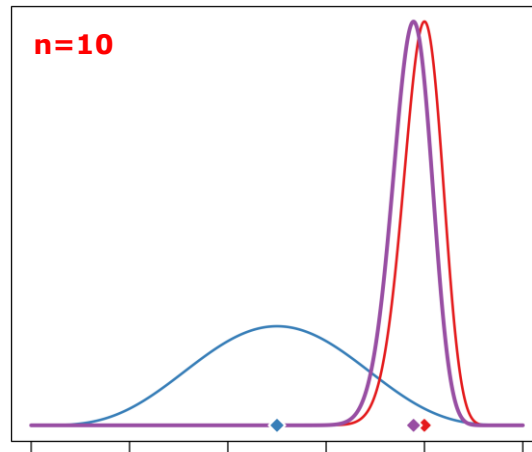
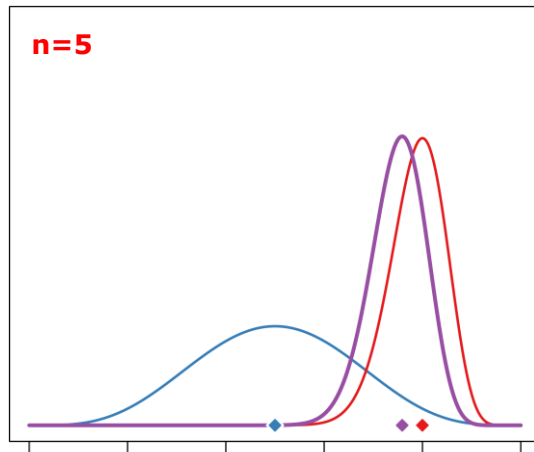
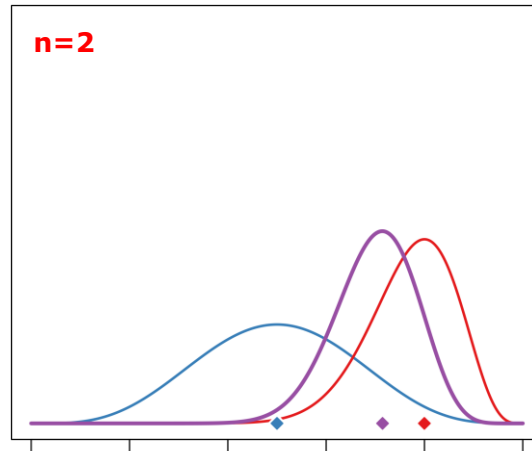
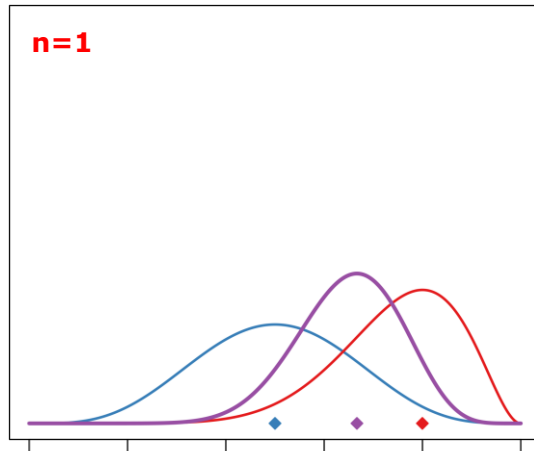
- Example: survival rate $\theta \in [0,1]$
- 1 Observation: 8/10 survived
- Binomial likelihood function
- Priors all beta distributions with mean = 0.5
- but different standard deviations (concentration around mean)

Prior affects the posterior

- For flat priors, posterior is proportional to likelihood
- For any other prior, posterior is a compromise between prior and likelihood
- For weakly informative priors, even little data dominates the posterior
- More informative priors (lower sdev) draw the posterior mean further away from the maximum likelihood estimate (MLE) towards the prior mean



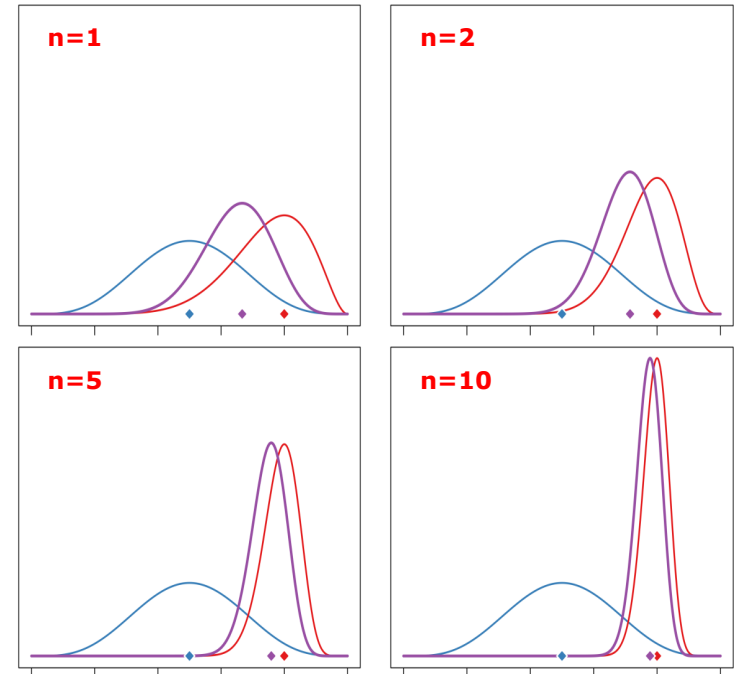
Likelihood affects the posterior



- Example: survival rate $\theta \in [0,1]$
- „informative prior“ from last slide
- Different numbers of obs. n
- Width of likelihood function decreases with n
(higher certainty)

Likelihood affects the posterior

- For small datasets (little experimental evidence), the prior can dominate the posterior
- In large datasets, likelihood can dominate the posterior
- Number of observations decreases the width of the likelihood and therefore also posterior uncertainty (stronger experimental evidence)
- Number of observations draws posterior mean towards maximum likelihood estimate (MLE)



Are priors subjective?

- Not including ANY information at all is also a choice
- Usually, you have SOME idea about a relationship
- If you have previous results (other studies / experiments), this information should go into your analysis
- Whole research process is never purely objective anyway
- In complex models, priors might even be necessary!
- If you're worried, perform a sensitivity analysis: re-analyze the data with different prior specifications



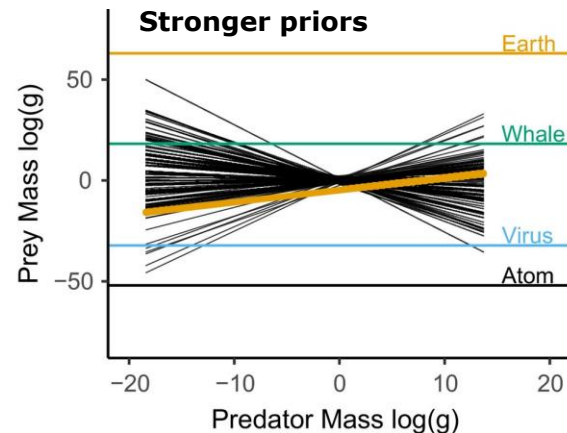
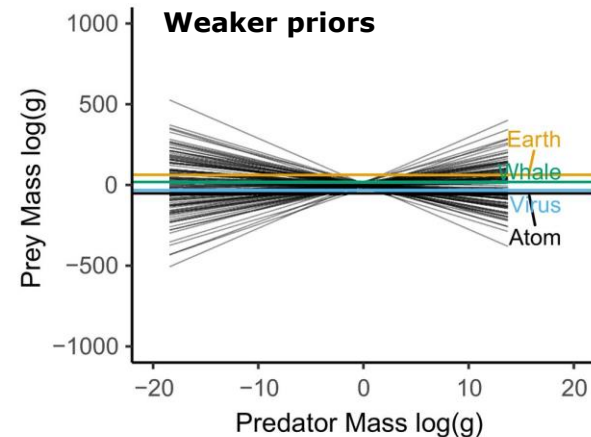
Prior predictive checks

- Test if priors make sense
- Generate predictions with samples from prior distribution
- Compare them to the range of observed data
- Helpful when using data transformations

(GLMs use nonlinear link-functions, like $\log$ or logit)

- *Traditional* viewpoint (old school):
Priors should be chosen before even looking at the data
- *Modern* viewpoint: Prior predictive simulations are useful!
E.g. McElreath: Statistical Rethinking (2020, 2nd ed.)

```
> prior_sim = brm(size~age, data=my_data, prior=my_prior,  
                  sample_prior="only")  
> pp_check(prior_sim)
```



brms default priors

- brms automatically chooses priors for **intercepts** and **standard deviations**

- Based on the observed data

- Overriding intercept default prior must be handled carefully:

brms internally uses mean-centered predictors, which changes intercept

→ My advice: leave them unless you want to include specific information on these parameters

- brms default priors for effect sizes / regression **slopes** are flat priors!

→ Choose your own priors for them!

```
> prior_summary(fit2)
```

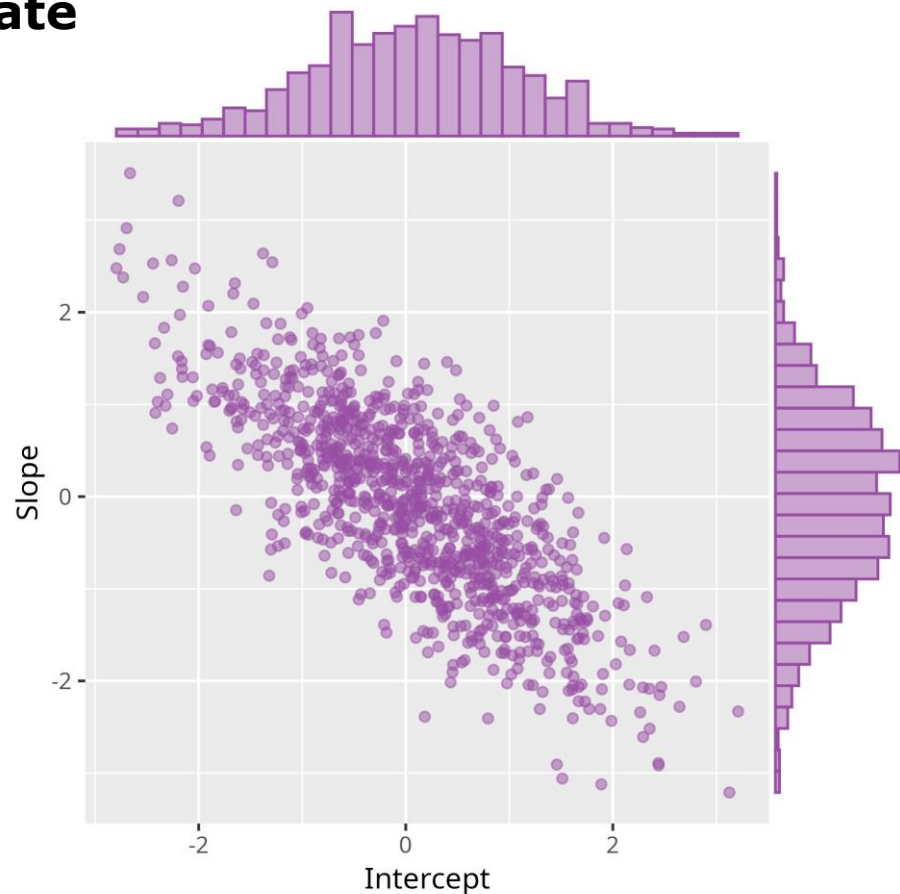
	prior	class	coef	group	resp	dpar	nlpar	lb	ub	source
	(flat)		b							default
	(flat)		b	age						(vectorized)
	student_t(3, 110, 11.9)	Intercept								default
	student_t(3, 0, 11.9)	sigma						0		default

Posterior distribution

The posterior sample is multivariate

	Intercept	Slope
[1,]	-0.524	0.638
[2,]	-0.703	0.473
[3,]	-0.200	0.942
[4,]	0.610	-0.357
[5,]	0.928	-0.581
[6,]	0.807	-0.941
[7,]	1.905	-1.924
[8,]	-0.136	-0.079
[9,]	-0.799	0.403
[10,]	0.098	-1.532

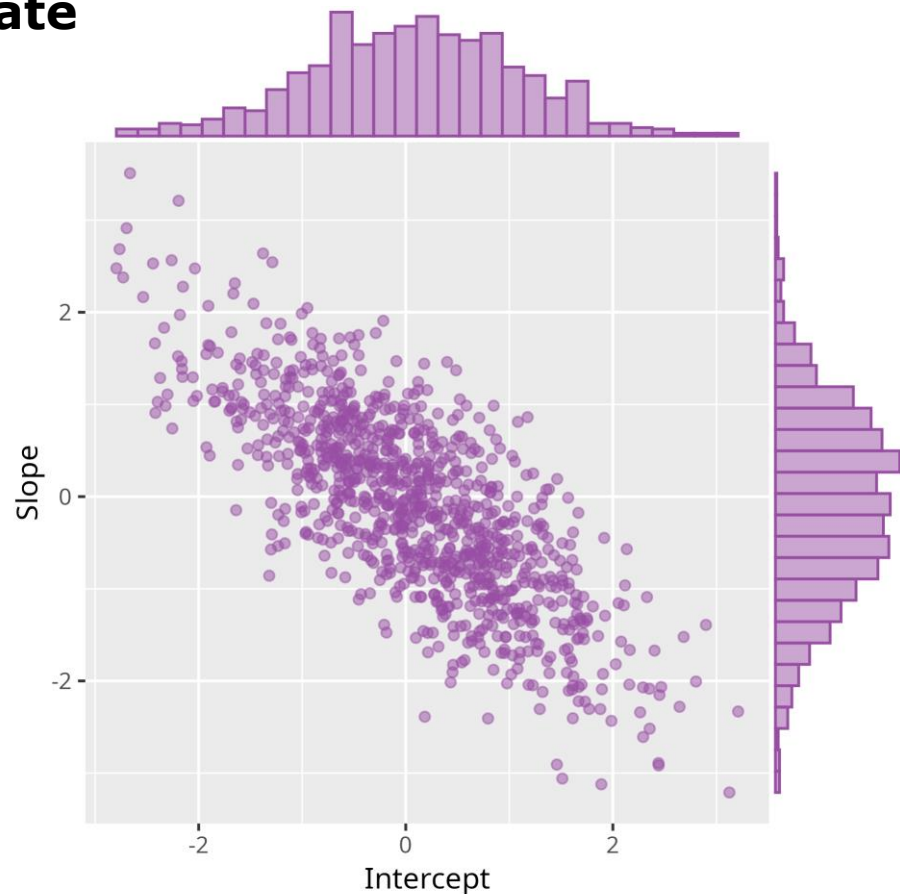
MCMC output is a matrix / dataframe !



The posterior sample is multivariate

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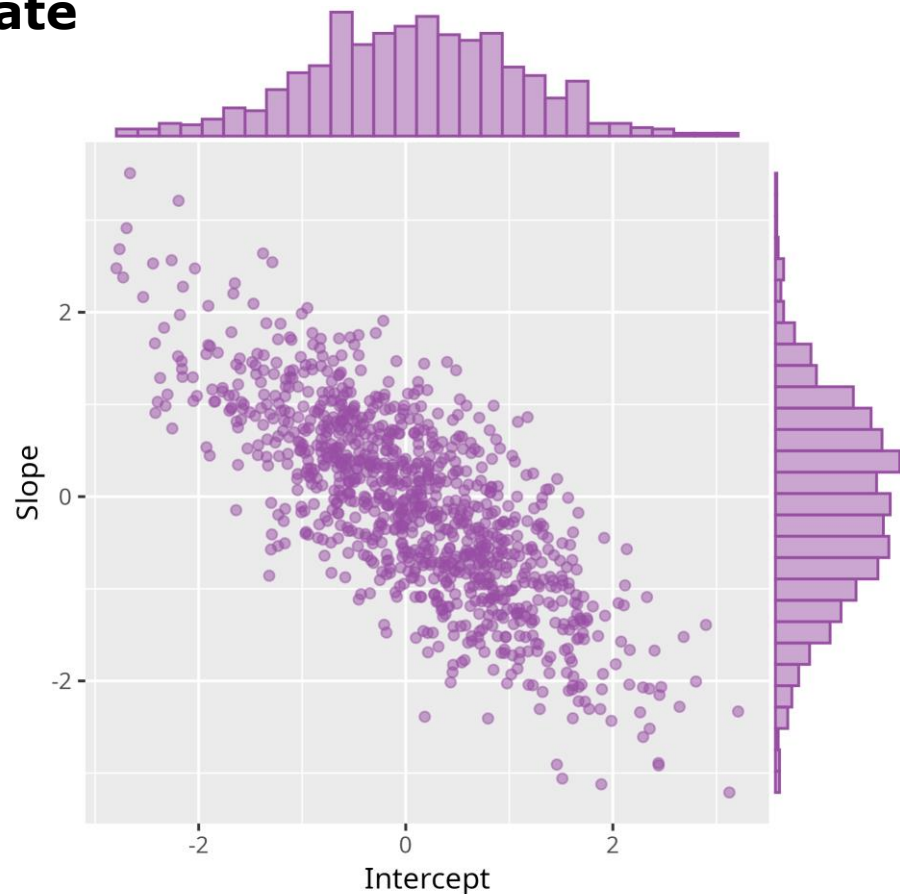
Each **column** contains all samples of 1 parameter



The posterior sample is multivariate

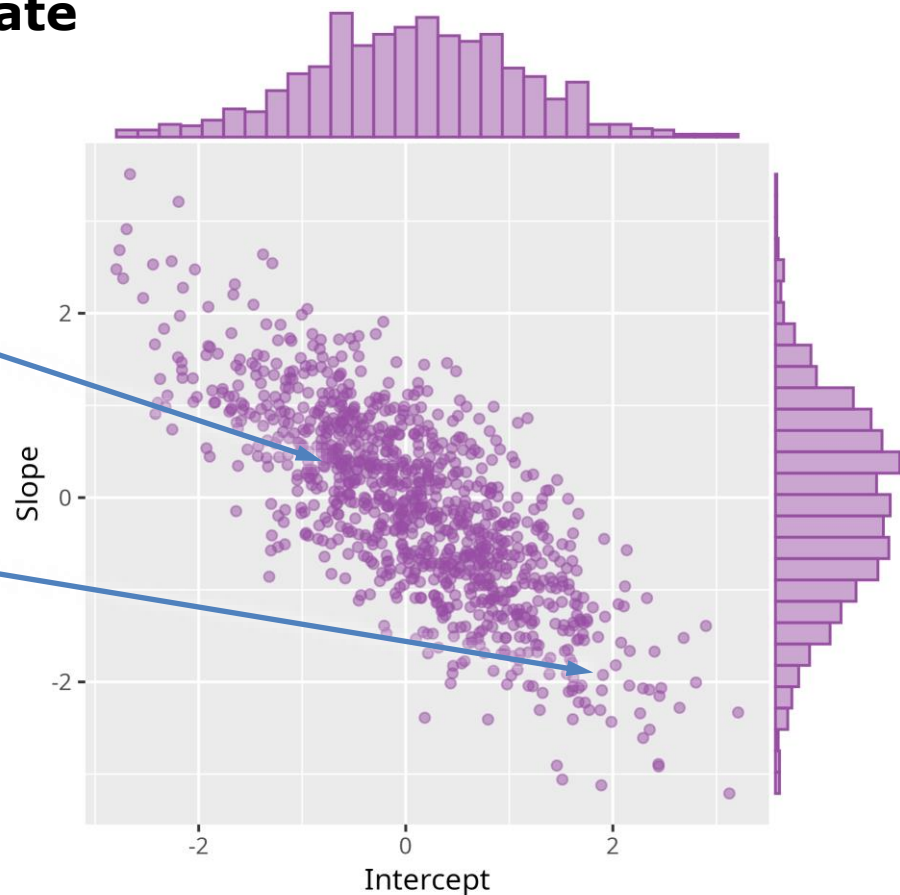
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[10,]	0.098	-1.532

Each **row** contains 1 sample of all parameters



The posterior sample is multivariate

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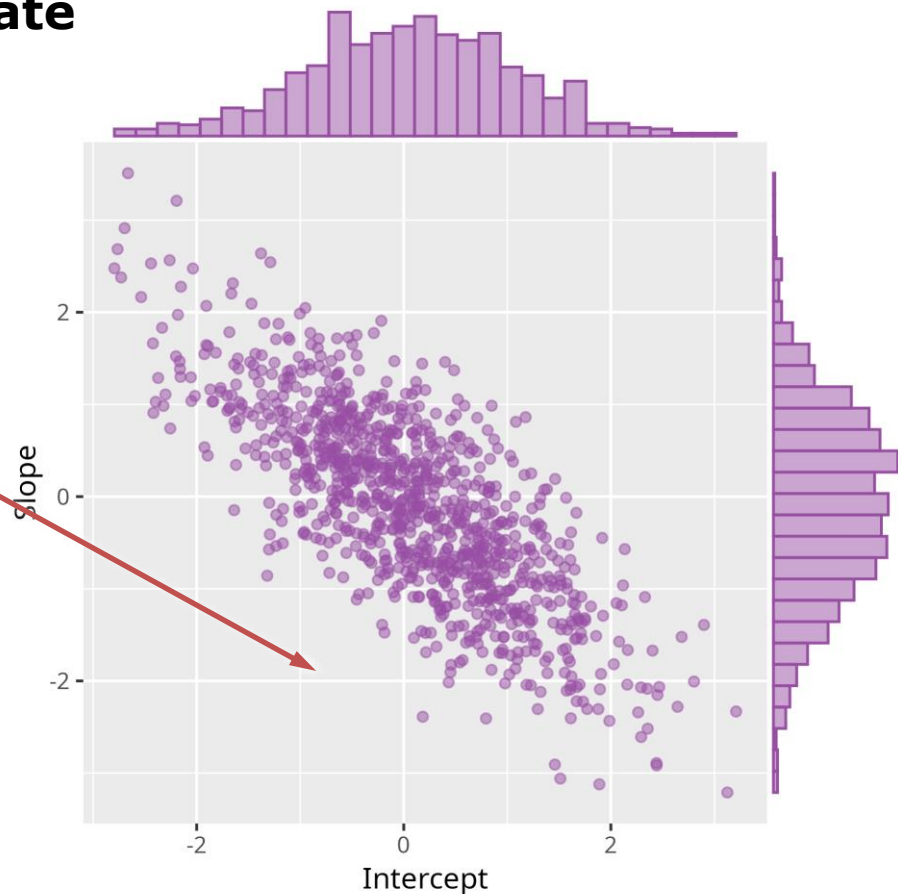


A random row is part of the posterior sample

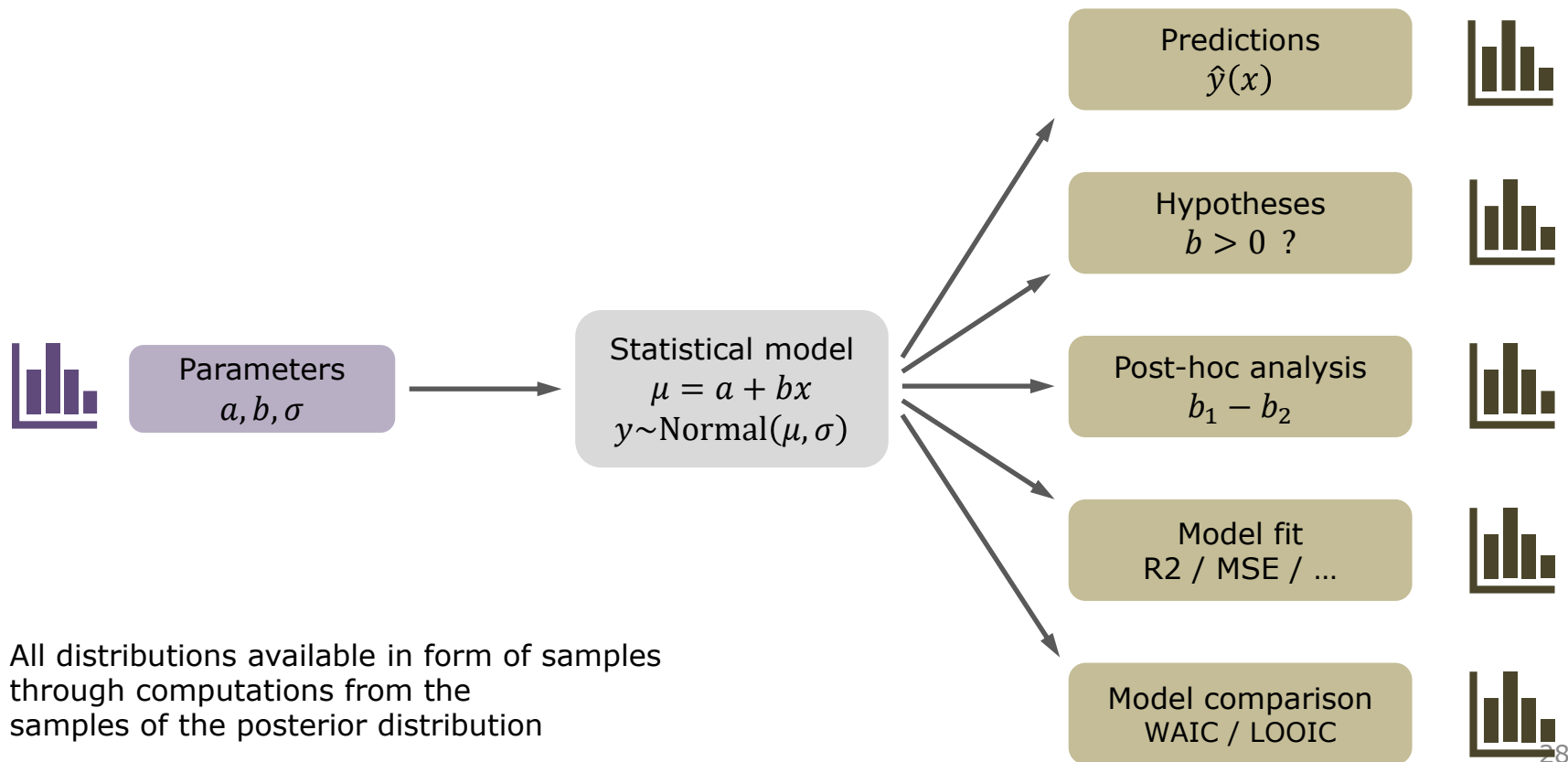
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Random entries of each column (mixed) are **not** part of the posterior sample



Everything is a distribution !



Posterior predictions

Example:

Example: linear relationship between age x and body mass y of sea turtles

Deterministic part:

$$\mu(x) = a + b \cdot x$$

Stochastic part:

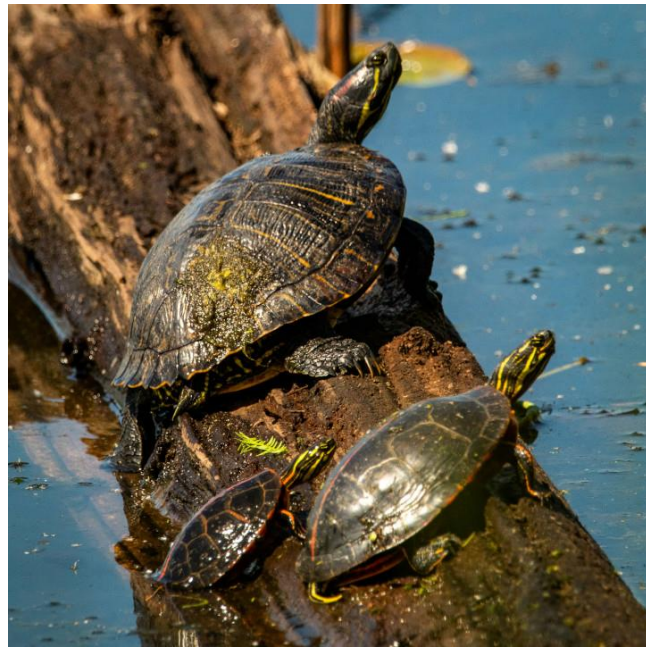
$$y \sim \text{Normal}(\mu, \sigma)$$

Parameters:

a intercept

b slope

σ standard deviation



Model output

```
> fit1 = brm(weight ~ age, data=data)
> summary(fit1)
```

Family: gaussian
Links: mu = identity; sigma = identity
Formula: weight ~ age
Data: data (Number of observations: 8)
Draws: 4 chains, each with iter = 2000; warmup = 1000; thin = 1;
total post-warmup draws = 4000

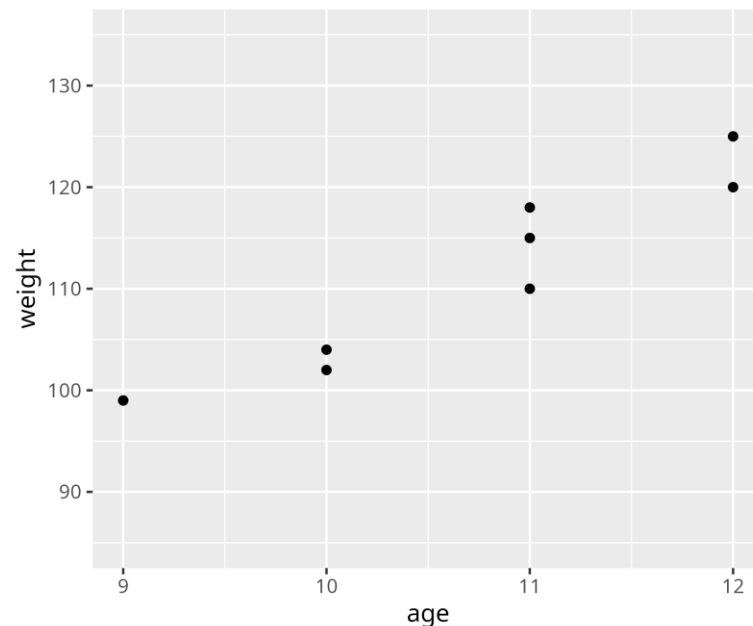
Regression Coefficients:

	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
Intercept	19.99	16.24	-13.79	53.23	1.00	2848	1916
age	8.53	1.50	5.46	11.59	1.00	2817	1838

Further Distributional Parameters:

	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
sigma	4.01	1.43	2.19	7.73	1.00	1606	1766

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS and Tail_ESS are effective sample size measures, and Rhat is the potential scale reduction factor on split chains (at convergence, Rhat = 1).



mean and sdev of parameters'
posterior distribution

How to predict ?

What is the predicted weight at age $x = 10.5$?

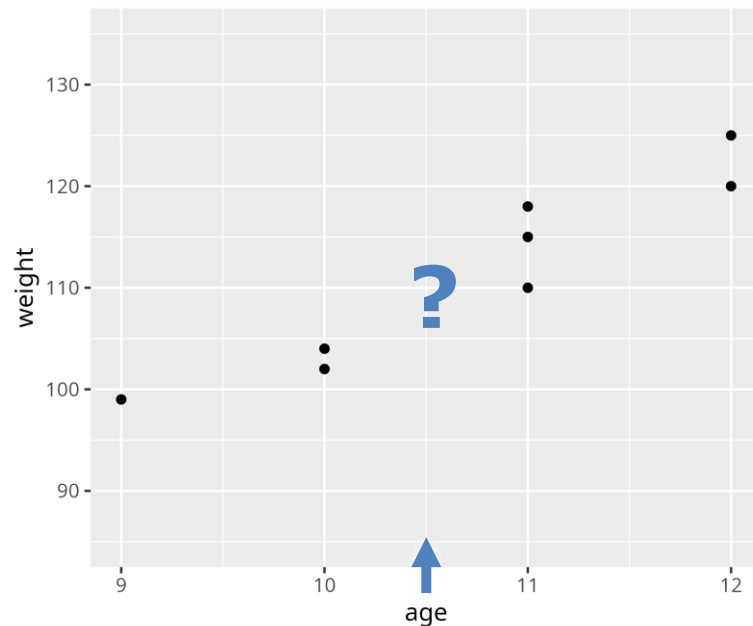
Deterministic part: $\mu(x) = a + b \cdot x$

We have mean values for

intercept $a = 19.99$ and slope $b = 8.53$

However, in Bayesian statistics, we **don't** use mean parameter values to make prediction.

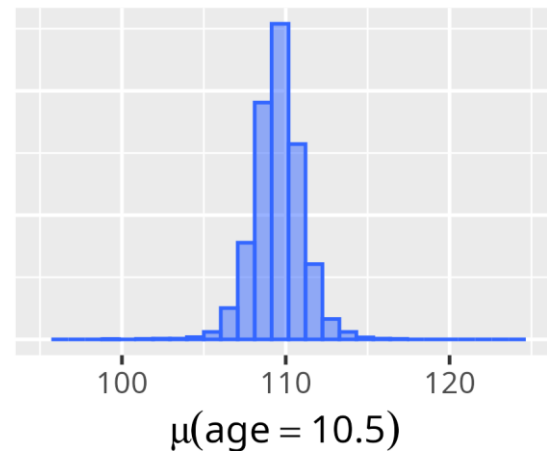
We use the **whole posterior distribution** to quantify prediction uncertainty correctly !



The fitted distribution

draw	b_Intercept	b_age	fitted(age=10.5)
1	7.473	9.738	109.720
2	30.477	7.614	110.425
3	22.153	8.273	109.022
4	34.008	7.310	110.763
5	39.144	6.667	109.147
6	12.472	9.328	110.415
7	7.005	9.674	108.579
8	-3.196	10.633	108.446
9	23.362	8.309	110.602
10	23.745	8.247	110.342

Deterministic part
 $\mu = a + b \cdot x$

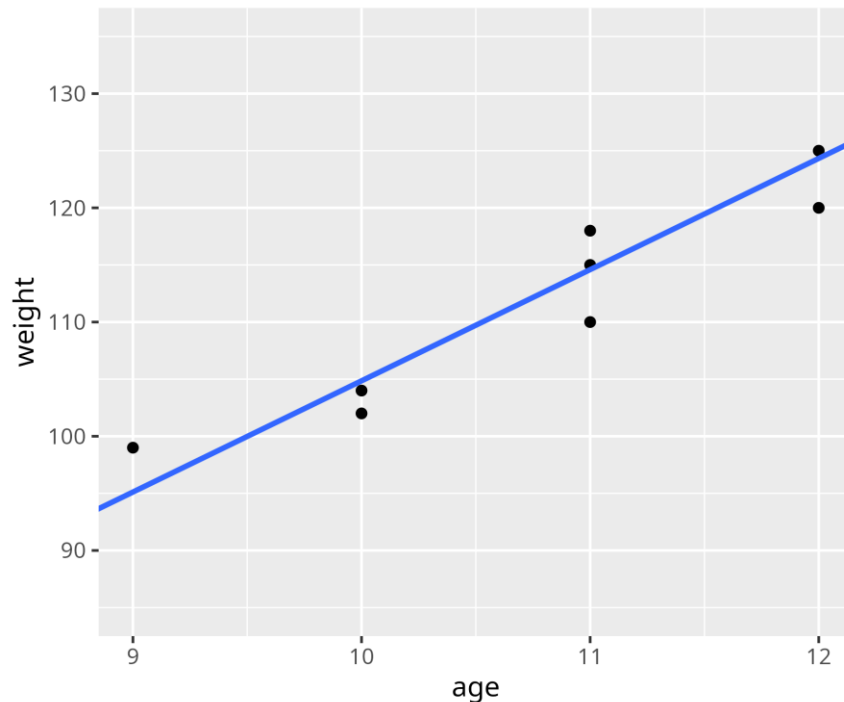


Each sample from the posterior (a_i, b_i) generates 1 sample for, $\mu_i(\text{age} = 10.5) = a_i + b_i \cdot 10.5$

```
> posterior_epred(fit1, newdata=data.frame(age=10.5))
```

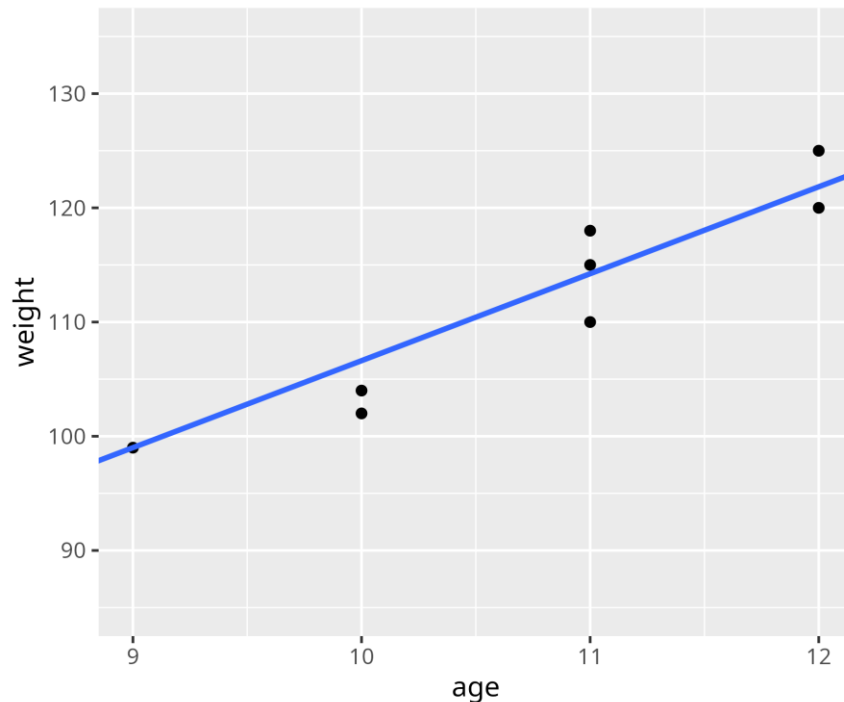
Each posterior sample generates a regression line

draw	b_Intercept	b_age	sigma
1	7.473	9.738	4.128
2	30.477	7.614	3.456
3	22.153	8.273	3.474
4	34.008	7.310	4.200
5	39.144	6.667	4.456
6	12.472	9.328	3.671
7	7.005	9.674	4.045
8	-3.196	10.633	4.117
9	23.362	8.309	3.519
10	23.745	8.247	3.336
11	27.194	7.948	4.249
12	18.373	8.566	3.351
13	20.358	8.522	3.447
14	10.307	9.322	3.007
15	24.486	8.224	3.408



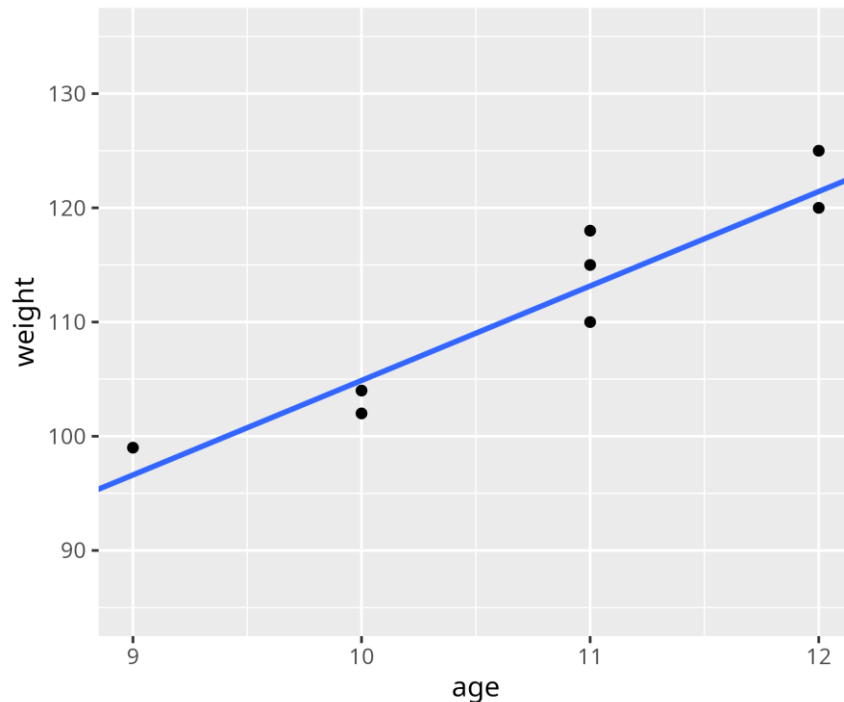
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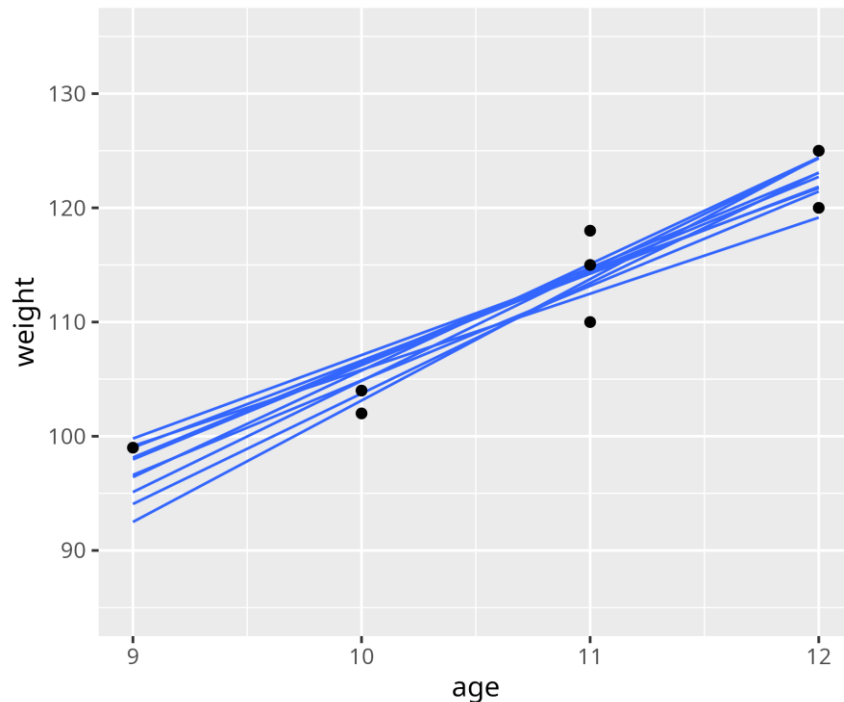
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Credible intervals

Distribution of fitted values / regression lines
with **deterministic model part only**

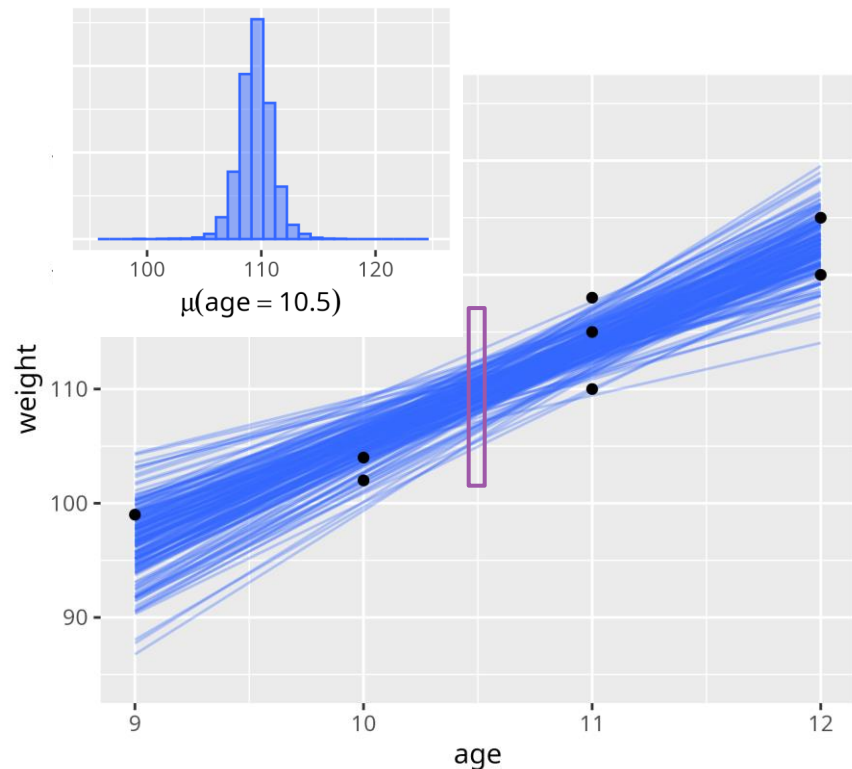
$$\mu_i(x) = a_i + b_i \cdot x \quad (i = 1, \dots, 1000)$$

Mean fitted value

$$\bar{\mu}(x) = \text{mean}(\mu_1(x), \dots, \mu_{1000}(x))$$

95% intervals are called **credible intervals**.
They quantify uncertainty of the regression line.

There is nothing magical about 95%,
can also choose other intervals, e.g. 90%



```
> plot(conditional_effects(fit1, spaghetti=TRUE, ndraws=200))
```

Credible intervals

Distribution of fitted values / regression lines
with **deterministic model part only**

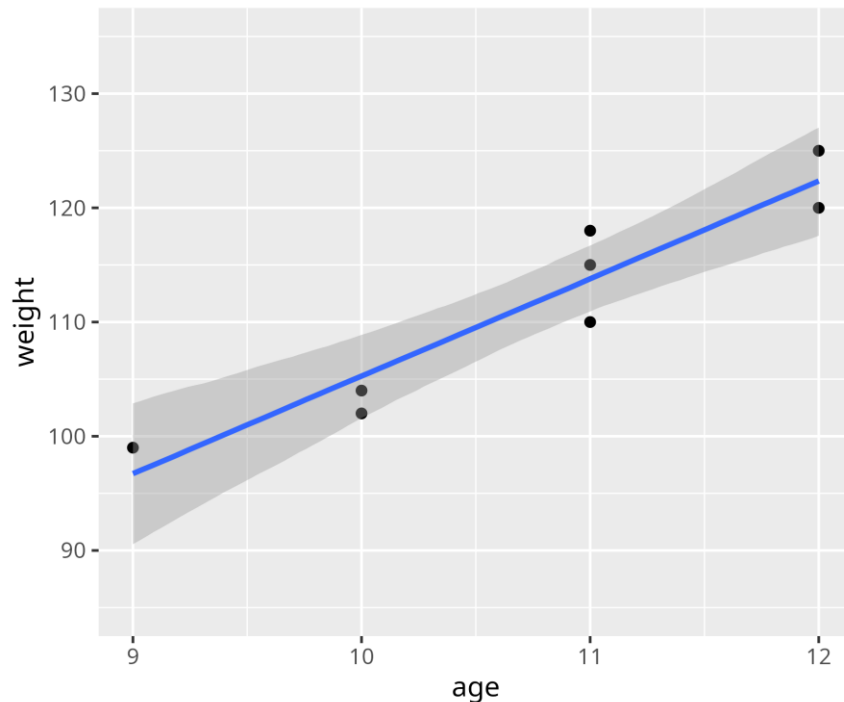
$$\mu_i(x) = a_i + b_i \cdot x \quad (i = 1, \dots, 1000)$$

Mean fitted value

$$\bar{\mu}(x) = \text{mean}(\mu_1(x), \dots, \mu_{1000}(x))$$

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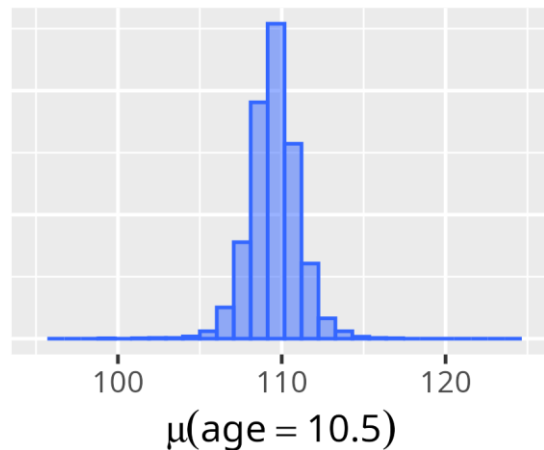


```
> plot(conditional_effects(fit1), points=TRUE)
```

The predictive distribution

b_Intercept	b_age	fitted(age=10.5)
7.473	9.738	109.720
30.477	7.614	110.425
22.153	8.273	109.022
34.008	7.310	110.763
39.144	6.667	109.147
12.472	9.328	110.415
7.005	9.674	108.579
-3.196	10.633	
23.362	8.309	
23.745	8.247	

Deterministic part
 $\mu = a + b \cdot x$

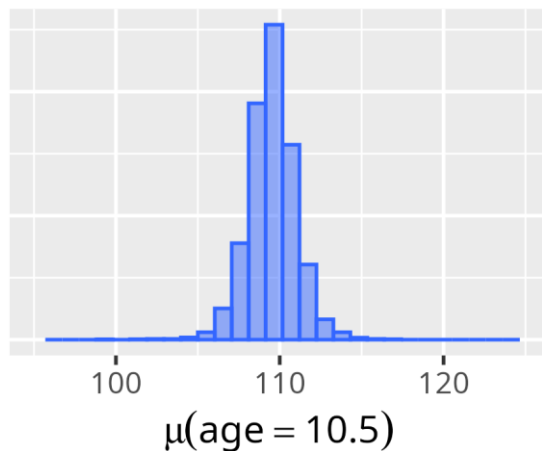


```
> posterior_epred(fit1, newdata=data.frame(age=10.5))
```

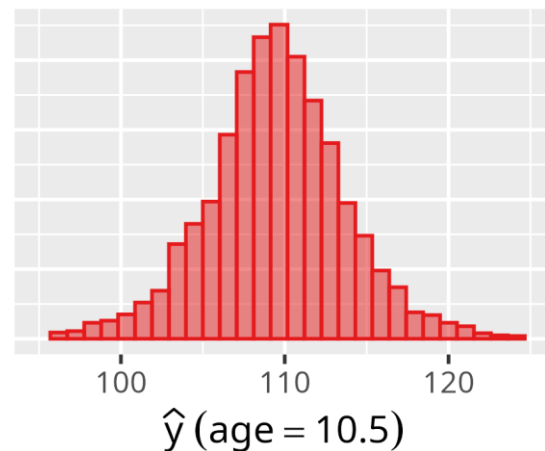

The predictive distribution

b_Intercept	b_age	fitted(age=10.5)	sigma	predict(age=10.5)
7.473	9.738	109.720	4.128	119.538
30.477	7.614	110.425	3.456	105.864
22.153	8.273	109.022	3.474	110.969
34.008	7.310	110.763	4.200	111.090
39.144	6.667	109.147	4.456	110.522
12.472	9.328	110.415	3.671	109.970
7.005	9.674	108.579	4.045	115.610
-3.196	10.633		4.117	
23.362	8.309		3.519	
23.745	8.247		3.336	

Deterministic part
 $\mu = a + b \cdot x$



Stochastic part
 $y \sim \text{Normal}(\mu, \sigma)$



> posterior_epred(fit1, newdata=data.frame(age=10.5))

> posterior_predict(fit1, newdata=data.frame(age=10.5))

Prediction intervals

Predictions add random residual error to fitted values

Distribution of predicted values

with **deterministic and stochastic model part**

$$\hat{y}_i(x) = \mu_i(x) + \varepsilon_i \quad (i = 1, \dots, 1000)$$

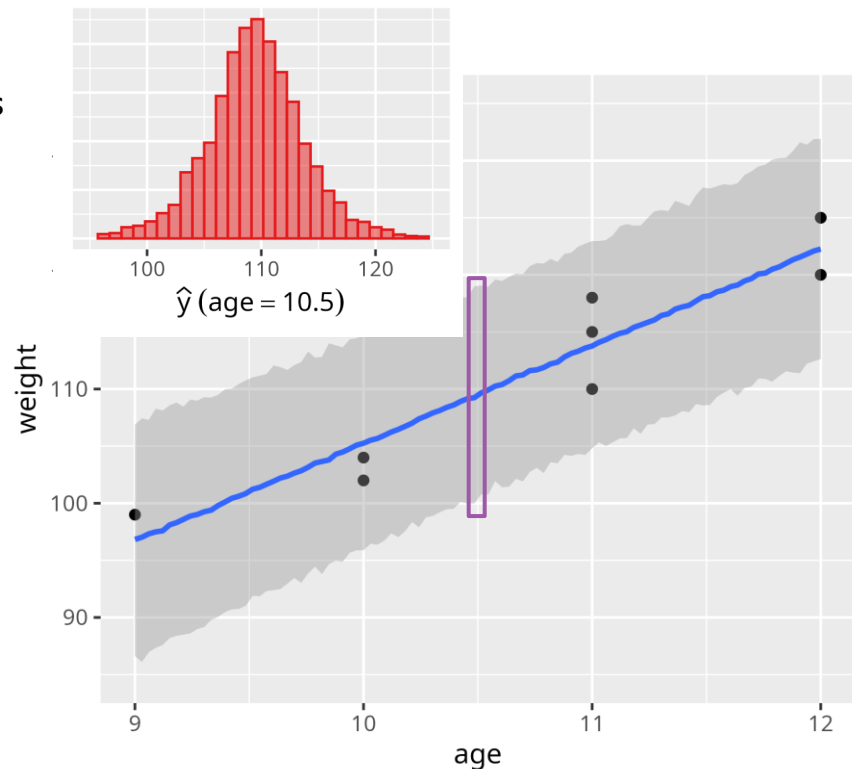
$$\varepsilon_i \sim \text{Normal}(0, \sigma_i)$$

Same as: $\hat{y}_i(x) \sim \text{Normal}(\mu_i(x), \sigma_i)$

95% intervals are called **prediction intervals**.

They quantify uncertainty of newly predicted data.

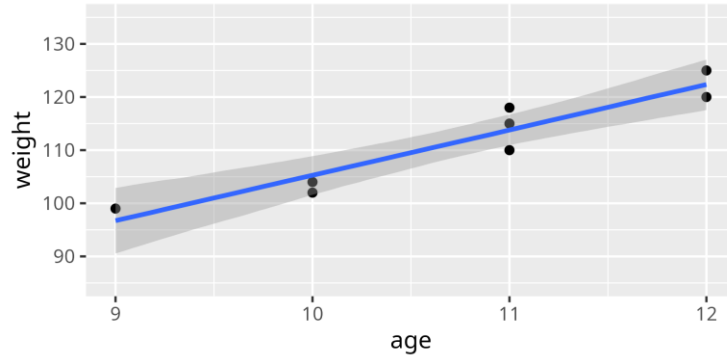
(Should contain around 95% of observed data.)



```
> plot(conditional_effects(fit1, method="posterior_predict"),  
       points=TRUE)
```

Fitted vs. predictive

Fitted

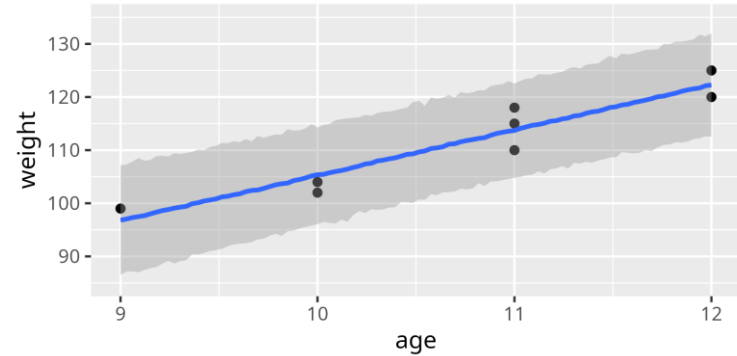


Mean regression line / curve under parameter uncertainty

„Credible intervals“

Uses deterministic model part only

Predicted



Predictive data distribution under parameter uncertainty and model residuals

„Prediction intervals“

Uses deterministic and stochastic model parts

Fitted vs. predictive

Fitted

```
> fitted(fit1)
```

	Estimate	Est.Error	Q2.5	Q97.5
[1,]	105.25324	1.813620	101.59634	108.8769
[2,]	122.30603	2.343272	117.55827	127.0220
[3,]	113.77964	1.458628	110.92357	116.6928
[4,]	113.77964	1.458628	110.92357	116.6928
[5,]	96.72685	2.995963	90.55468	102.8774
[6,]	113.77964	1.458628	110.92357	116.6928
[7,]	105.25324	1.813620	101.59634	108.8769
[8,]	122.30603	2.343272	117.55827	127.0220

Mean regression line / curve under parameter uncertainty

„Credible intervals“

Uses deterministic model part only

Predicted

```
> predict(fit1)
```

	Estimate	Est.Error	Q2.5	Q97.5
[1,]	105.22283	4.549012	95.94294	114.3628
[2,]	122.18553	4.887509	112.25245	131.7162
[3,]	113.76152	4.490723	104.56416	122.8968
[4,]	113.70727	4.512039	104.60886	122.9418
[5,]	96.82354	5.324598	86.61397	107.5571
[6,]	113.75514	4.443002	104.59945	122.6224
[7,]	105.24951	4.562843	95.97297	114.4820
[8,]	122.42399	4.799248	112.77401	132.2151

Predictive data distribution under parameter uncertainty and model residuals

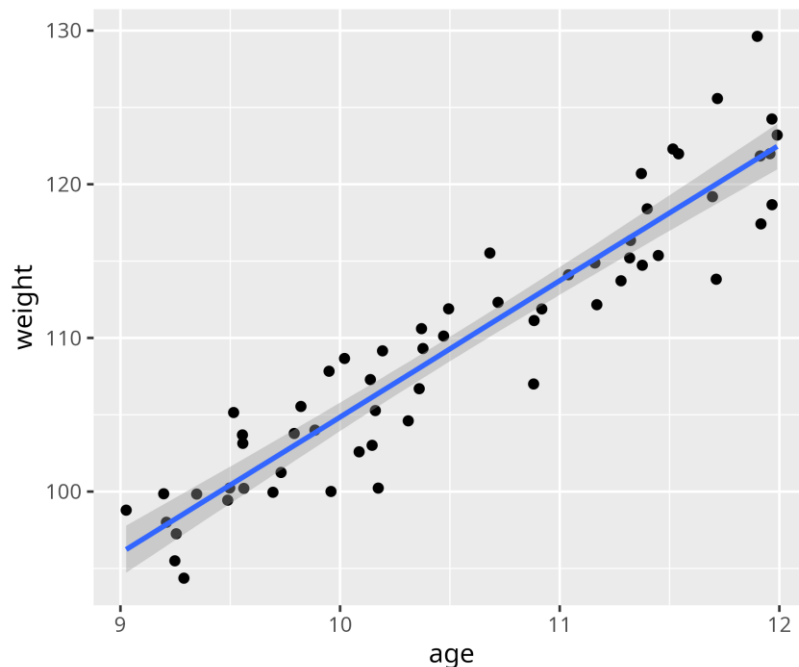
„Prediction intervals“

Uses deterministic and stochastic model parts

Posterior predictive checks

Linear regression assumptions

- 1. Linearity:** Trend of y follows linear prediction model $\mu(x) = a + b \cdot x$
- 2. Independence** of residuals:
not correlated (e.g. space or time)
- 3. Equal variance** of residuals:
across whole range of x
- 4. Normality** of residuals:
 $\varepsilon \sim \text{Normal}(0, \sigma)$



Model checking

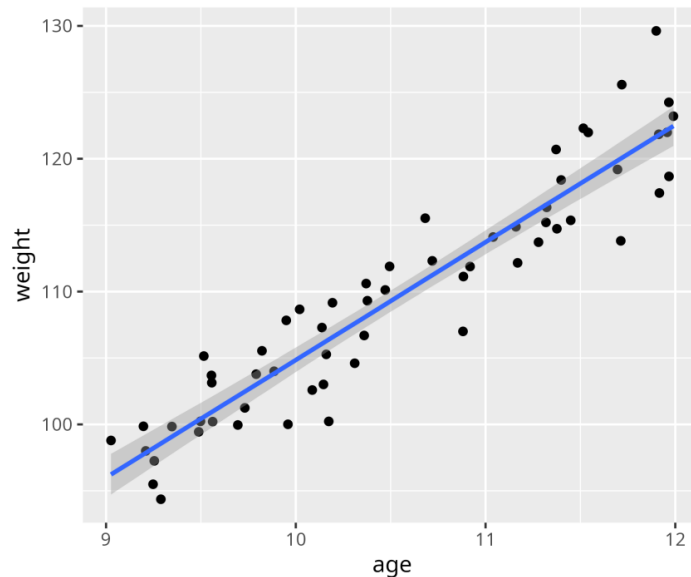
Visualization is easy when you have just one predictor!

Need alternative visual tools when dealing with multiple predictors.

Response / prediction is just 1 variable

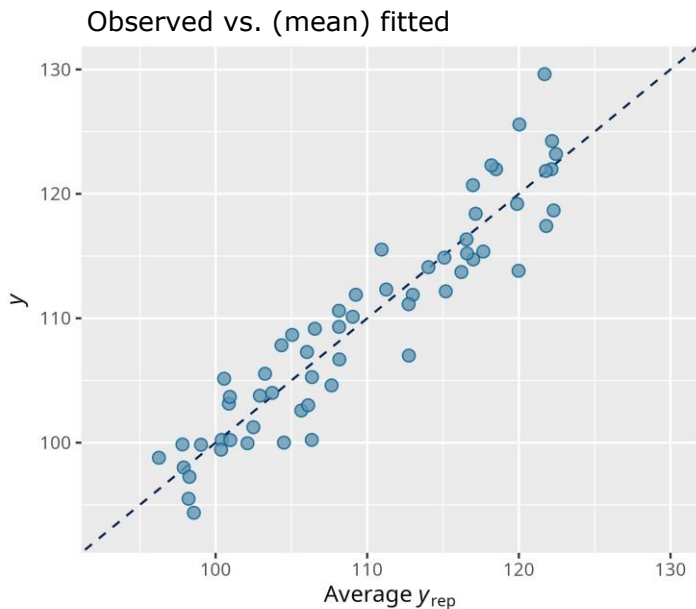
→ Compare and plot against each other:

- observations
- (mean) predictions
- residuals (observed - predicted)

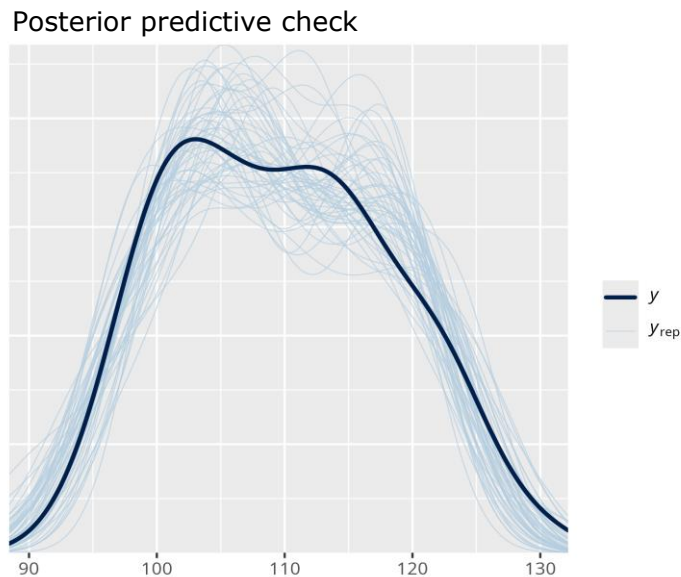


```
> plot(conditional_effects(fit1), points=TRUE)
```

Model checking (from brms package)



```
> pp_check(fit1, type="scatter_avg")
```

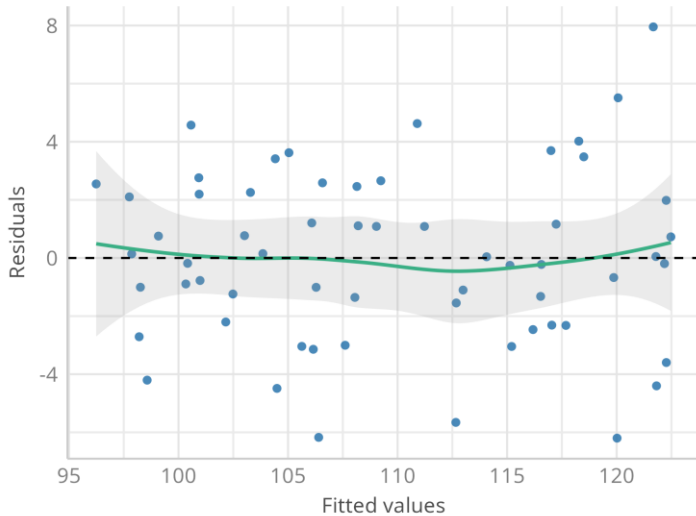


```
> pp_check(fit1, ndraws=50)
```


Model checking (from performance package)

Linearity

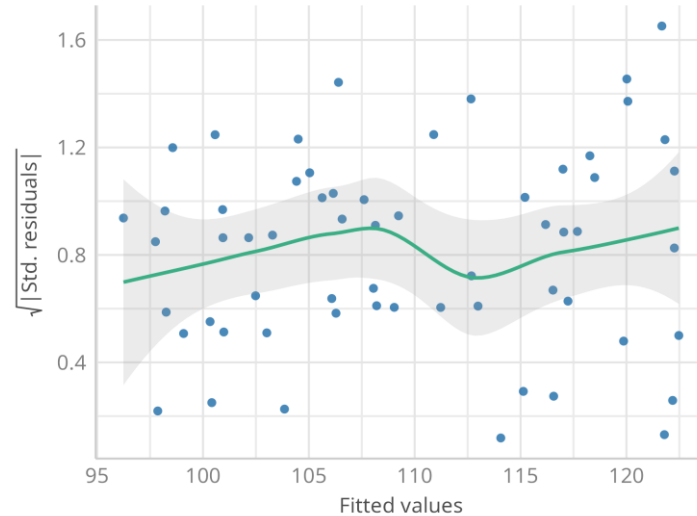
Reference line should be flat and horizontal



```
> check_model(fit1, check=„linearity“)
```

Homogeneity of Variance

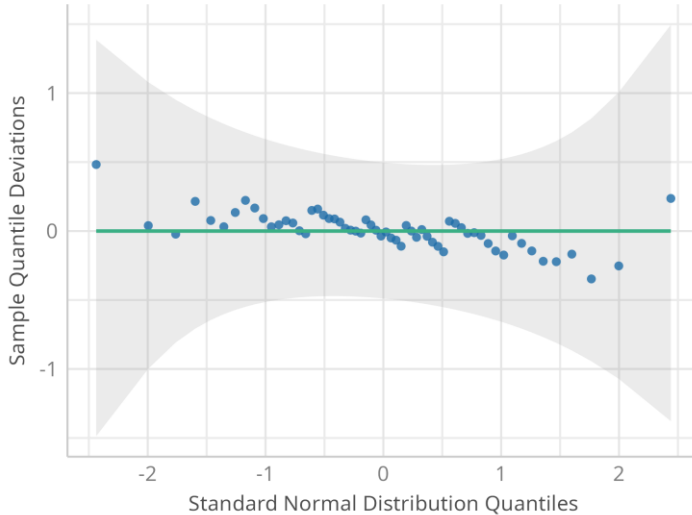
Reference line should be flat and horizontal



```
> check_model(fit1, check=„homogeneity“)
```

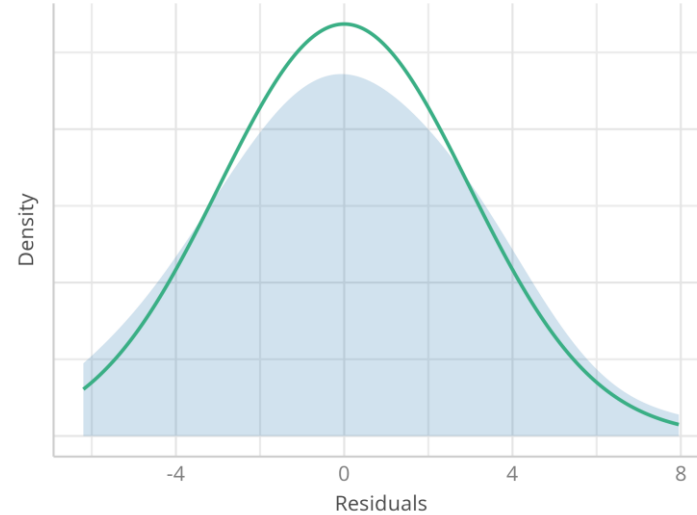
Model checking (from performance package)

Normality of Residuals
Dots should fall along the line



```
> check_model(fit1, check=„qq“)
```

Normality of Residuals
Distribution should be close to the normal curve



```
> check_model(fit1, check=„normality“)
```

Pitfalls of prediction: Multivariate posterior

If your posterior (parameters a, b) was shaped like Croatia, (nonlinear correlation), then the mean $(\bar{a}, \bar{b})$ in 2d-space would not be part of the posterior sample

Parameter combination $(\bar{a}, \bar{b})$ is **highly unlikely**

Prediction $\mu(\bar{a}, \bar{b})$ is not the mean prediction, but rather meaningless!

→ **Always use full posterior for making predictions!**



Due to its shape, the centre of Croatia is actually located in Bosnia and Herzegovina

[Post übersetzen](#)



Pitfalls of prediction: Jensen's inequality

For linear models μ (as in linear regression $\mu = a + bx$)

$$\text{mean}(\mu(\theta_1), \dots, \mu(\theta_{1000})) = \mu(\text{mean}(\theta_1, \dots, \theta_{1000}))$$

$\underbrace{\hspace{10em}}$

Distribution of
predictions

$\underbrace{\hspace{10em}}$

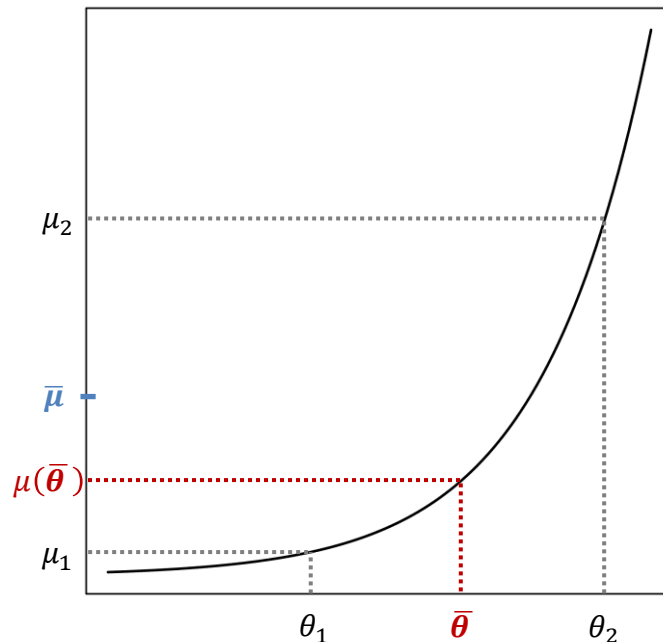
Prediction with
point estimate $\bar{\theta}$

For **nonlinear functions** μ

$$\text{mean}(\mu(\theta_1), \dots, \mu(\theta_{1000})) \neq \mu(\text{mean}(\theta_1, \dots, \theta_{1000}))$$

Relevant for nonlinear statistical models, GLMs,
link functions, parameter or data transformations

→ **Always use full posterior for making predictions!**



Summary

Summary

- Priors → you choose !
 - Likelihood → given by data & statistical model
 - MCMC samples from posterior → **check convergence** !
 - Informative priors can decrease uncertainty in posterior
 - More datapoints can decrease uncertainty in posterior
 - Use posterior predictions to **check model assumptions and model fit**
 - In Bayesian statistics, everything is a distribution
- Use full posterior (samples) for everything

Further reading

Banner, K. M., Irvine, K. M., & Rodhouse, T. J. (2020). The use of Bayesian priors in Ecology: The good, the bad and the not great. *Methods in Ecology and Evolution*, 11(8), 882–889. <https://doi.org/10.1111/2041-210X.13407>

Bürkner, P. (2024). The brms Book [in progress]. <https://paulbuerkner.com/software/brms-book/>

Conn, P. B., Johnson, D. S., Williams, P. J., Melin, S. R., & Hooten, M. B. (2018). A guide to Bayesian model checking for ecologists. *Ecological Monographs*, 88(4), 526–542. <https://doi.org/10.1002/ecm.1314>

Gabry, J., Simpson, D., Vehtari, A., Betancourt, M., & Gelman, A. (2019). Visualization in Bayesian workflow. *Journal of the Royal Statistical Society. Series A, (Statistics in Society)*, 182(2), 389–402. <https://doi.org/10.1111/rssa.12378>

Lemoine, N. P. (2019). Moving beyond noninformative priors: why and how to choose weakly informative priors in Bayesian analyses. *Oikos*, 128(7), 912–928. <https://doi.org/10.1111/oik.05985>

McElreath, R. (2020). Statistical Rethinking: A Bayesian Course with Examples in R and STAN (2nd ed.). *Chapman and Hall/CRC*. <https://doi.org/10.1201/9780429029608>

van de Schoot, R., Depaoli, S., King, R., et al. (2021). Bayesian statistics and modelling. *Nature Reviews. Methods Primers*, 1(1), 1–26. <https://doi.org/10.1038/s43586-020-00001-2>

Wesner, J. S., & Pomeranz, J. P. F. (2021). Choosing priors in Bayesian ecological models by simulating from the prior predictive distribution. *Ecosphere*, 12(9), e03739. <https://doi.org/10.1002/ecs2.3739>

<https://github.com/stan-dev/stan/wiki/prior-choice-recommendations>